Lineare Algebra
Serie 26
Lukas Mengis

1.

Let $v, w \in V$ and b symmetric bilinear from of V
with:

$$q_B(v) = B(v,v) = \langle v\ ,\ v \rangle$$

Since B is linear and bilinear

$$\begin{aligned} B(w+v, w+v) &= \langle v\ ,\ w+v \rangle + \langle w\ ,\ w+v \rangle \\ &= \langle v\ ,\ v \rangle + \langle v\ ,\ w \rangle + \langle w\ ,\ w \rangle + \langle w\ ,\ v \rangle \\ &= q_B(v) + q_B(w) + 2B(v,w) \end{aligned}$$

$$\Rightarrow B(v,w) = \frac{1}{2}(q_B(v+w) - q_B(v) - q_B(w)).$$

2.

2.a.

Let A be a real Matrix

$$A := \frac{1}{3}\begin{pmatrix} 2 & -2 & 1 \\ -1 & -2 & -2 \\ 2 & 1 & -2 \end{pmatrix} = \begin{pmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{pmatrix}$$

$$\frac{\left|\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}\right|}{3} = \frac{\sqrt{4+1+4}}{3} = 1$$

$$\frac{\left|\begin{pmatrix} -2 \\ -2 \\ 1 \end{pmatrix}\right|}{3} = \frac{\sqrt{4+4+1}}{3} = 1$$

$$\frac{\left|\begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix}\right|}{3} = \frac{\sqrt{1+4+4}}{3} = 1$$

$$\langle v_1\ ,\ v_2 \rangle = \frac{-4+2+2}{9} = 0$$

$$\langle v_1\ ,\ v_3 \rangle = \frac{2+2-4}{9} = 0$$

$$\langle v_2\ ,\ v_3 \rangle = \frac{-2+4-2}{9} = 0$$

$\Rightarrow v_1, v_2, v_3$ form an orthonormal Basis. Therefore, $\det(A) = 1$ and A is orthogonal

2.b.

We want to find the fixed points

$$
\begin{aligned}
Av &= v \\
\Rightarrow (A - I)v &= 0 \\
&= \frac{1}{3}\begin{pmatrix} -1 & -2 & 1 \\ -1 & -5 & -2 \\ 2 & 1 & -5 \end{pmatrix} v = 0
\end{aligned}
$$

$$
\Rightarrow v \in \operatorname{span}\left(\begin{pmatrix} -3 \\ 1 \\ -1 \end{pmatrix}\right)
$$

$\Rightarrow \operatorname{span}(v)$ is the rotational axis. Set $w = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

$$
v \cdot w = \begin{pmatrix} -3 \\ 1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = 0 + 1 - 1 = 0
$$

$$
\begin{aligned}
\cos(\varphi) &= \frac{w \cdot Aw}{||w|| \cdot ||Aw||} \\
&= \frac{-1 - 4}{\sqrt{2}\sqrt{18}}
\end{aligned}
$$

$$
\Rightarrow \varphi = \cos^{-1}\left(-\frac{5}{6}\right)
$$

3.

3.a.

$$
\det(3) = 3 > 0
$$

$$
\det\left(\begin{pmatrix} 3 & 3 \\ 3 & 1 \end{pmatrix}\right) = -6 < 0
$$

A is not positive definite

3.b.

$$
\det(6) = 6 > 0
$$

$$
\det\begin{pmatrix} 6 & 3 \\ 3 & 7 \end{pmatrix} = 33 > 0
$$

$$
\det\begin{pmatrix} 6 & 3 & 4 \\ 3 & 7 & 3 \\ 4 & 3 & 8 \end{pmatrix} = 170 > 0
$$

B is positive definite

3.c.

maybe, maybe not

4.

4.a.

Let $A \in M_{n\times n}(\mathbb{R})$ be a symmetric matrix and A is positive definite.
$\forall v \in \mathbb{R}^n \setminus \{0\}$

$(A) \Rightarrow (B)$
Let λ be an eigenvalue for v

$$Av = \lambda v$$

A is positive definite

$$v^T A v = v^T \lambda v = ||v||^2 \lambda > 0$$
$$||v||^2 > 0 \Rightarrow \lambda > 0$$

$(B) \Rightarrow (C)$ Since A is symmetric $\Rightarrow$ A is orthogonal diagonalizable

$$A = P^T D P$$

with $D = \mathrm{diag}(\lambda_1, ..., \lambda_n)$. A and D have the same eigenvalues (all greater than 0)

$$D = S^2 = \mathrm{diag}\left(\sqrt{\lambda_1}, ..., \sqrt{\lambda_n}\right)^2 = \mathrm{diag}(\lambda_1, ..., \lambda_n)$$

Since every eigenvalue is greater than 0, the square root of the eigenvalues is in $\mathbb{R}$

$$A = P^T D P = P^T S^2 P = P^T S P P^T S P = \left(P^T S P\right)^2$$

$\left(P^T S P\right)$ is symmetric and invertible since S is invertible and P to.
$(C) \Rightarrow (A)$

$$S^2 = A$$

$$v^T A v = v^T S^2 v = v^T S^T S v = ||Sv||^2 > 0$$

5.

5.a.

f does a combination of rotation and reflection

$$f : V \to V$$
$$x \mapsto Rx, \quad \text{with} \quad R \text{ orthogonal Matrix}$$

Every row and column has length 1

$$\Rightarrow \left|r_{i,j}\right| \leq 1$$
$$\mathrm{Tr}(f) = \sum_{i=1}^{n} r_{ii} \leq \sum_{i=1}^{n} 1 = n$$
$$\mathrm{Tr}(f) = \sum_{i=1}^{n} r_{ii} \geq \sum_{i=1}^{n} -1 = -n$$
$$\Rightarrow |\mathrm{Tr}(f)| \leq n$$

5.b.

$|\mathrm{Tr}(f)| = n \Rightarrow R = I$ or $R = -I$

6.

Two 2-dimesional subspaces $E_1, E_2 \subset \mathbb{R}^3$
Set E_1 and E_2 equal to the span of each two orthonormal vectors.

$$E_1 = \mathrm{span}(e_1, e_2) \quad E_2 = \mathrm{span}(e_1', e_2')$$

Since the basis vectors are orthonormal there exists an orthogonal endomorphism R from one set of basis vectors to the others.

$$R\begin{pmatrix} | & | \\ e_1 & e_2 \\ | & | \end{pmatrix} = \begin{pmatrix} | & | \\ e_1' & e_2' \\ | & | \end{pmatrix}$$

$$\Rightarrow RE_1 = E_2$$

Add to the bases a third vector such that $\operatorname{span}(e_1, e_2, e_3) = \operatorname{span}(e_1', e_2', e_3') = \mathbb{R}^3$. with

$$e_3 \coloneqq e_1 \times e_2 \qquad e_3' \coloneqq e_1' \times e_2'$$

$T = R$

$$T_1(e_1, e_2, e_3) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} e_1' \\ e_2' \\ e_3' \end{pmatrix}$$

or

$$T_2(e_1, e_2, e_3) = \begin{pmatrix} \cos(\theta) & \sin(\theta) & 0 \\ \sin(\theta) & -\cos(\theta) & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} e_1' \\ e_2' \\ e_3' \end{pmatrix}$$

$\theta \in [0, 2\pi)$

$$\Rightarrow T_1(E_1) = T_2(E_1) = E_2$$